

IoT Deployment Templates

Deployment templates provide standard architectures for implementing IoT systems.

Template 1: Single Device Deployment

Components:

- One sensor/actuator
- Microcontroller
- User interface

Example: Temperature monitoring system using Arduino.

Flow:

Sensor → Arduino → User

Template 2: Single Device with Cloud

Components:

- Sensor
- Microcontroller
- Internet Connection
- Cloud Platform

Example: Weather monitoring station.

Flow:

Sensor → Controller → Cloud → User

Template 3: Multiple Devices with Gateway

Components:

- Multiple sensors
- Gateway
- Cloud server

Example: Smart home system.

Flow:

Sensors → Gateway → Cloud → User

Template 4: Multiple Devices with Edge Analytics

Components:

- Sensors
- Edge Gateway
- Cloud

Features:

- Local processing
- Reduced latency
- Faster response

Example: Industrial automation systems.

Template 5: Enterprise Deployment

Components:

- Large number of devices
- Multiple gateways
- Cloud infrastructure
- Analytics engine

Example: Smart city solutions.

Flow:

Devices → Gateways → Cloud → Analytics → Applications